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Tendon-driven continuum robots with extensible sections—A model-based evaluation of path-following motions

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Abstract

Continuum robots are highly miniaturizable, exhibit non-linear shapes with several curves, and are flexible and compliant. In particular, concentric-tube and tendon-driven continuum robots can be designed on a small scale with diameters of below 10 mm. A small diameter-to-length ratio enables insertion of these robots through small entry points in order to reach hardly accessible regions by avoiding obstacles. This scenario can often be found in minimally invasive surgery and technical inspections. However, to reach the target region, a deployment along a narrow tortuous path is often required. Common tendon-driven continuum robots are intrinsically incapable of such deployment and concentric-tube continuum robots require special path conditions and intensive parameter optimization. Other proposed robot types, such as hyper-redundant and pneumatically actuated robots, exhibit less favorable diameter-to-length ratios and are thus not suitable for those tasks. Since the limiting factors are found in the design of continuum robots, we propose a novel tendon-driven continuum robot design, which features an additional degree of freedom in each robot section. The backbone is composed of straight, concentrically arranged tubes, each of which composes a section and is used to adapt its length. We present a three-section continuum robot prototype with a diameter of 7 mm, determine its follow-the-leader capabilities theoretically, and validate the results experimentally using model-based control. For our 165 mm long robot prototype, the repeatability is below 2.38 mm. The model accuracy reaches a median of 3.16 % over 25 configurations with respect to robot length. The path-following error over five curvilinear paths results in median errors of 2.59 % with respect to robot length.

Keywords

Continuum robots, tendon-driven, follow-the-leader, model-based control, tendon elongation modeling

1. Introduction

In comparison with conventional robots, continuum robots are not composed of joints and rigid links, but of one or several elastic structures which can bend and twist or elongate, or both, resulting in a curvilinear shape of the structure and smooth motions. The slender continuous structure enables obstacle avoidance and therefore increases accessibility of hardly reachable regions in confined spaces. Being elastic, continuum robots are compliant and can adjust to the environment under external loads without causing any damage to the environment or themselves. In general, the structure of a continuum robot foresees a concatenation of several independent sections, which can be deformed individually. Diverse continuum robot designs have been proposed thus far. Recent overviews were provided by Burgner-Kahrs et al. (2015) and Walker (2013).

In this paper, we are particularly concerned with accessing workspaces that are characterized by tortuous paths in highly constrained environments, such as minimally invasive surgery or industrial inspection tasks. These require continuum robots with a small diameter-to-length ratio, which are able to access the environment and can be deployed along tortuous paths to reach the region of interest. In the following, we review the related state of the art in continuum robotics, focused on those robot designs that allow for deployment, feature extension, or contraction, and are able to achieve path-following motions.

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1.1. Related work

A common continuum robot design, which is intrinsically actuated, features pneumatic or hydraulic chambers. Grouping three or four chambers in parallel constitutes a continuum section, which allows for bending in two planes by inflating and deflating some of the chambers. Elongation or contraction can be achieved by inflating or deflating all chambers simultaneously. By arranging the chambers helically, twisting can also be realized. The Octarm uses parallel McKibben actuators in three sections (Bartow et al., 2014) with section diameters of 64–90 mm. The current version achieves controllable section length ranges of 0.44–0.4 m, 0.47–0.41 m, and 0.57–0.5 m. Ranzani et al. (2015) introduced a soft continuum robot with a diameter of 32 mm using parallel fluidic actuators and granular jamming to achieve stiffening of individual sections. For a two-section robot, an elongation up to 62 % could be achieved. Maghooa et al. (2015) proposed a combination of tendon and pressure actuation and realized a prototype with 40 mm diameter at the base, decreasing to 10 mm at the tip. While the robot exhibits considerable contraction and elongation, quantitative evaluation is not provided. To summarize, continuum robots that rely on intrinsic actuation by hydraulics or pneumatics, usually provide three degrees of freedom per section, such that path-following motion can theoretically be achieved as the curvature and the length can be actuated simultaneously. As the lower bound of contraction is defined by the initial length and the maximum elongation by the length of the fully pressurized pneumatic or hydraulic chambers, the range of length per section is limited. For deployment along a tortuous path, the range of section lengths should be as wide as possible, ideally allowing for zero section length, so that a wide range of bending radii can be realized.

Extrinsically actuated continuum robots allow for small diameter-to-length ratios. For instance, tendon-driven continuum manipulators (e.g. in Gravagne et al. (2003), Simaan et al. (2009)) consist of an elastic backbone equipped with a number of spacer disks fixed to the backbone equidistantly. A number of tendons are routed in parallel to the backbone through the spacer disks. Tendons are fixed to an end disk defining the end of a section. Instead of tendons, push–pull rods can also be used (Simaan et al., 2009). The curvature of each section can be varied by pushing or pulling on the tendons, but the section length is fixed. With three or four tendons per section, bending in two planes can be achieved, such that one section features two degrees of freedom. To deploy such a tendon-driven continuum robot along a tortuous path, an additional translational degree of freedom has to be realized, e.g. by mounting the whole robot on a linear stage. Such a setup is associated with large path deviation errors (up to 25 mm, i.e. 17% with respect to length) as soon as the path consists of two or more different curvatures, as shown in our previous work (Amanov et al., 2017). A tendon-driven continuum robot design with eight sections and 24 degrees of freedom for aerospace turbine inspection was investigated for path-following motion by Palmer et al. (2014). Large path deviations, of at least 25 mm, were shown in experiments. Although adding an arbitrary routed tendon offers additional degrees of freedom and increases the robot's dexterity (Gao et al., 2015; Starke et al., 2017), it cannot compensate for the fixed section length.

Another extrinsically actuated mechanism can be found in concentric-tube continuum robots. These are composed of telescoping precurved elastic tubes, which are nested inside one another (Dupont et al., 2010; Webster III et al., 2009). Each tube can be rotated and translated individually. Through elastic interaction between the tubes, the overall shape of the continuum robot conforms to several pieces of constant curvature, which can be considered as individual sections. As a result, section lengths and curvatures depend on each other such that they cannot necessarily be adjusted individually while preserving the other sections' lengths and curvatures. The range of curvatures is further limited and determined by the tube design (precurvatures). This can be a limiting factor, if concentric-tube robots are considered for operation in hardly accessible environments that require deployment along tortuous paths. In fact, path-following motion can only be achieved with very limited curvatures and actuation requirements (Gilbert and Webster III, 2013) and necessitates an application-based design parameter optimization, as well as explicit path-planning techniques (Granna et al., 2016). Yet, in terms of miniaturization, concentric-tube continuum robots show the greatest potential among continuum robots (diameter below 2 mm).

To address the limitations in path-following capabilities, researchers proposed robot designs. Degani et al. (2006) investigated path-following deployment based on alternating deployment of two concentrically arranged snake robots with shape-lock mechanism. The robot is capable of following arbitrary paths. As the shape locking relies on friction between surfaces of the snake robot links, miniaturization in terms of diameter is limited (the presented prototype has a diameter of 15 mm). Kang et al. (2016) introduced a variable length push–pull rod continuum robot design consisting of two segments. The design intrinsically allows for path-following motion (as proven qualitatively), but also requires a diameter of 30 mm in the current prototype, as shape lockers based on piezoelectric actuators need to be integrated. Tappe et al. (2015) introduced a binary actuated hyper-redundant robot for path-following deployment and proved its capabilities in simulation. As the robot is composed of a series of electromagnetic tilting actuators, miniaturization is challenging. The diameter of the current robot (45 mm) is not favorable for highly constrained environments, such as accessing the human body through natural orifices, as reported by the authors. Approaches involving shaping a shaft into the desired path and presetting the deployment of a robot have been proposed. However, this approach requires a-priori guiding shaft fabrication and can only be used for one particular path (Henselmans et al.,

2017). To the best of our knowledge, a continuum robot design with diameter below 10 mm that can adjust its section curvature and length independently for deployment along tortuous paths has not been proposed thus far.

1.2. Contribution

In this paper, we build on our previous work toward a tendon-driven continuum robot design with extensible sections (Nguyen and Burgner-Kahrs, 2015). With three degrees of freedom per section (two for bending and one for extension or contraction), our robot design enables approximate path-following motion along paths composed of circular arcs. We have previously derived a kinematic model and proved the follow-the-leader capabilities in simulation (Neumann and Burgner-Kahrs, 2016).

The contribution of this paper is an updated and improved robot design with a motorized actuation unit. The new robot prototype exhibits three extensible sections with nine degrees of freedom and the actuation unit provides 12 controlled axes, enabling automatic control of the robot. We further extend our theoretical results by (1) analyzing the workspace of the robot and pointing out the increase compared with a conventional tendon-driven continuum robot and (2) path-following deployment using forces as model input. We provide an extensive experimental evaluation, investigating the repeatability and the model accuracy for the robot prototype. To improve the model accuracy, we derive a tendon elongation model, show the benefits of the derived method, and conduct quantitative path-following experimental investigations on a three-section robot.

2. Robot design

The general concept combines the designs of concentric-tube and tendon-driven continuum robots, which has been presented in preliminary form in Nguyen and Burgner-Kahrs (2015). The backbone is composed of n elastic, straight tubes arranged concentrically inside one another, such that they are able to slide along a common central axis in a telescopic manner. An end disk is rigidly attached to each tube's tip. Tendons attach to each end disk to induce bending. Each tendon is routed in parallel to the backbone through spacer disks. While conventional tendon-driven continuum robot designs feature spacer disks rigidly attached to the backbone with equidistant spacing, our concept foresees spacer disks equipped with permanent magnets, which distribute with equidistant spacing and thus adjust to the backbone length.

The length of a section l_i is controlled by extension and retraction of the ith backbone tube. Figure 1 illustrates a continuum robot with three sections, and shows the robot with maximum section lengths (Figure 1(b)), a fully retracted third section (Figure 1(c)), and fully retracted second and third sections (Figure 1(d)). The length of a section is defined as the distance between the end disks of two consecutive sections or the robot outlet. Each section length $l_i \in [l_{\min}, l_{\max}]$ is restricted to $l_{\max}$, depending on the number of spacer disks and the magnetic repulsion force. The minimum length $l_{\min}$ is determined by $l_{\min} = p \cdot h$, where p represents the number of spacer disks of the section and h the thickness of a spacer disk.

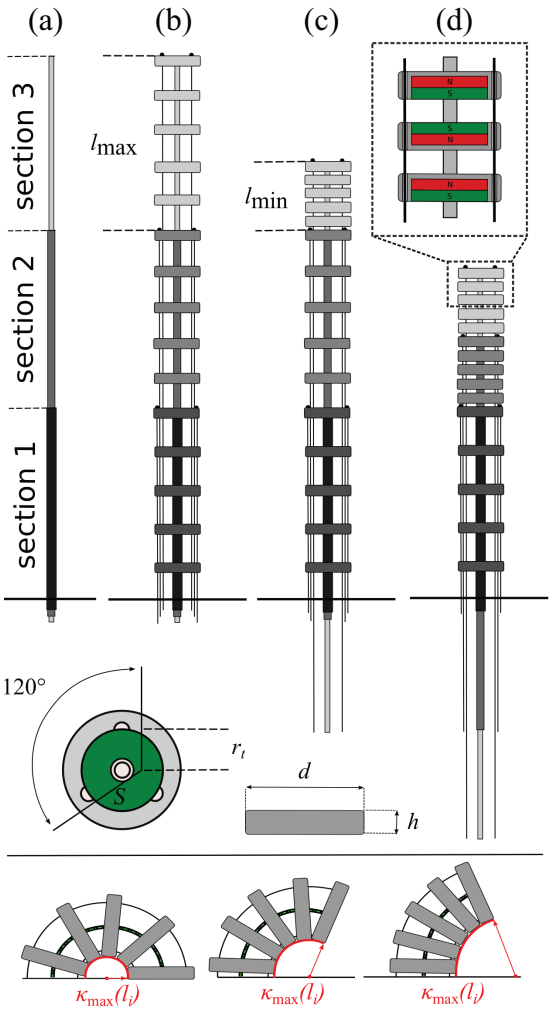


Fig. 1. Top: (a) Telescopic backbones composed of three straight, elastic tubes; (b) equipped with permanent magnet spacer disks at full length; (c) with the third section fully contracted; (d) with the third and second section fully contracted. Middle: Spacer disk geometry including the magnet tap and tendon-routing channels. Bottom: Maximum curvature dependency on the section length.

The curvature of each section of the continuum robot is controlled by tendon actuation. The number of tendons can be chosen according to the requirements of the application. Here, we use a variant with three circumferential tendon-routing channels with 120° offset (see Figure 1). The maximum curvature of each section is defined by the actual

Table 1. Tubular backbone parameters: outer tube diameter $\varnothing_o$, inner diameter $\varnothing_i$, and mass distribution m_{tube}. Robot prototype parameters: magnet radius r_t, maximum and minimum section length l_{max} and l_{min}, disk height h, disk diameter d, number of disks per section p, and disk weight m_{disk}.

Parameter		Tube 1		Tube 2		Tube 3	
$\varnothing_o$ (mm)		1		0.8		0.5	
$\varnothing_i$ (mm)		0.88		0.54		0.35	
m_{tube} (g/m)		3.5069		1.792		0.674	
Parameter	r_t (mm)	l_{max} (mm)	l_{min} (mm)	h (m)	d (m)	p	m_{disk} (g)
Value	2.5	55	15	1.5	7	10	0.2

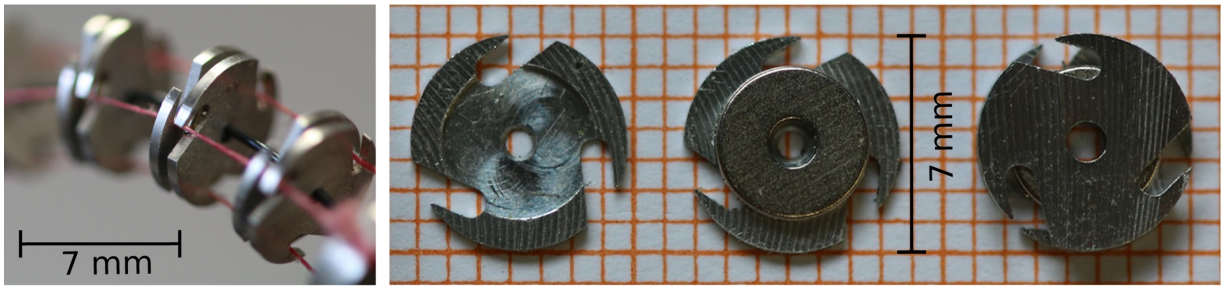


Fig. 2. Close-up of tendon-routing through a spacer disk (left). Parts for assembly of a spacer disk, with a permanent magnet sandwiched between two disk halves (right).

section length l_i and spacer disk radius $d/2$, as shown in Figure 1(bottom), and can be calculated by

$$\kappa_{max}(l_i) = \frac{l_i - l_{min}}{l_i \cdot d/2}. \quad (1)$$

The maximum recoverable strain of the backbone material ε determines the maximum curvature of the section

$$\kappa_\varepsilon = \frac{2\varepsilon}{\varnothing_o(1+\varepsilon)} \quad (2)$$

with $\varnothing_o$ the largest tube diameter of the respective backbone section. Thus, $\kappa_{max}(l_i) \leqslant \kappa_\varepsilon$ must be fulfilled to prevent permanent deformation of the section.

2.1. Prototype

The prototype used in this paper is composed of $i = 3$ extensible sections. The backbone tubes are made of NiTi in its austenite phase (recoverable strain ε of 8%). The tube's parameters are summarized in Table 1.

While each section's end disk is realized from a single cylindrical part, the spacer disks are composed of three parts: a permanent magnet sandwiched between two aluminum spacer disk halves (see Figure 2). This disk composition simplifies the robot assembly and tendon threading. We use circular N45 neodymium magnets with 5 mm diameter and 1 mm thickness. The magnet diameter determines the diameter of the spacer disks and the minimum routing channel distance to the center of the robot ($r_t = 2.5$ mm). The central circular holes of the spacer disks are 1.2 mm, 1 mm, and 0.7 mm, to comply with the backbone tube diameters. The choice of tendon material and parameters is discussed in the following sections.

The robot prototype is depicted in Figure 3. Our prototype allows for a maximum section length l_{max} of 55 mm. The minimum section length l_{min} is limited to 15 mm since the spacer disk height h is 1.5 mm and $p = 10$ disks (including the end disk). The maximum section length l_{max} can be up to 100 mm for 10 disks while ensuring equidistant distribution. However, we recommend that shorter lengths be chosen to provide enough repulsion force during simultaneous steering and elongation or contraction. While environmental contact can temporarily affect the disk distribution, equidistant spacing is recoverable.

2.2. Actuation unit

An actuation unit for the proposed robot needs to provide twelve controlled axes, nine for the tendons and three for the tube translation. We seek a compact and simple, yet effective, actuation unit design. Therefore, we use DC



Fig. 3. Three-section robot prototype in an example bent configuration.

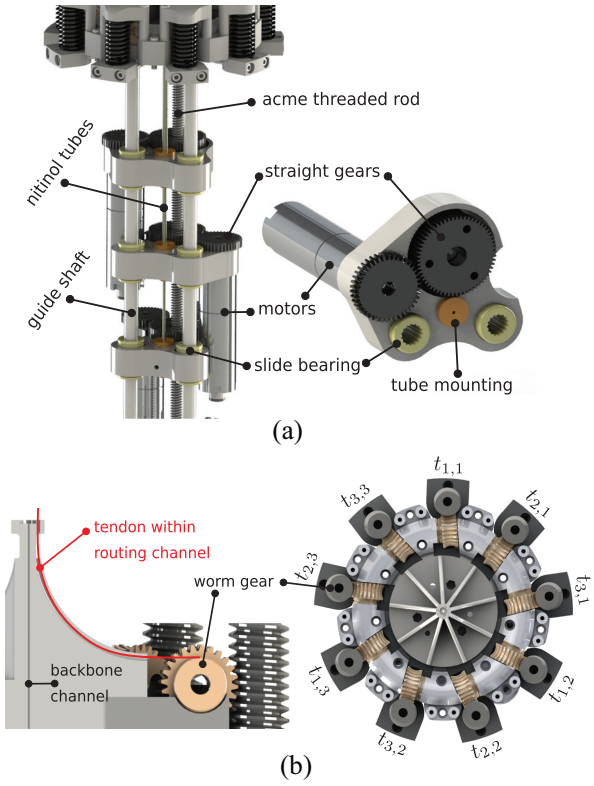


Fig. 4. (a) Tube actuation part with a close-up of a carriage. (b) Tendon-routing and tendon actuation part in cross-section and top view.

motors with integrated encoders for position control, and omit additional force sensors for tendon tension measurement. The actuation unit consists of a tube actuation part (Figure 4(a)), a tendon-routing part (Figure 4(b) left), and a tendon actuation part (Figure 4(b) right).

The tendon-routing part guides the tendons on a circular path to the tendon actuation, where they are actuated by DC motors (RE25, Maxon Motor AG, Sachseln, Switzerland) over a worm gear with a ratio of 20:1. The motors are equipped with encoders (1024 tics per revolution) and a gearbox with a ratio of 15:1. The worm wheels are mounted in the housing using slide bearings. The tendons are spooled on the worms in a 1 mm groove. For three sections, nine identical motor worm gear modules are arranged circumferentially around the robot's origin at equidistant angular displacements of 40°, as shown in Figure 4(b).

The tube actuation part consists of an acme threaded rod, a screw nut and a pair of gear wheels driven by DC motors (identical to those used for tendon actuation), as depicted in Figure 4(a). All these parts are mounted inside a carriage. Three linear guides per carriage ensure precise linear motion. The tubes are mounted inside the carriage within collets. Each carriage holds one tube. The motors are controlled by two motion control boards (DMC 4080 and DMC 4040, Galil Motion Control, Rocklin, CA, USA) realizing proportional–integral–derivative (PID) control.

2.3. Tendon selection

For the tendon selection, two requirements resulting from the robot design are important to consider. First, withstanding relatively high tendon forces, up to 30 N, for the first section (derived from kinematic model observations) while fitting through a 1 mm tendon-routing channel. Second, low tendon elongation in order to prevent undesired shape deviations. We are relying on tendon displacement feed-forward control to avoid addition of tension sensors. Although tendon displacement can be controlled by DC motors with feedback from the encoders, we cannot prevent tendon elongation related to tendon forces. To address the first requirement, we seek for tendons with high tensile strength as they need to withstand large loads at small diameters. To address the second requirement, we seek for tendons with large Young modulus (large force-to-elongation ratio), to reduce a possible shape error caused by tendon elongation under load. Braided threads consisting of ultra-high molecular weight polyethylene (UHMWPE) fibers, like Dyneema®, offer a suitable solution, since they withstand at least comparable or higher loads with the same diameter than steel (high tensile strength), and show little elongation (high Young modulus). Additionally, UHMWPE fibers are very flexible and can only be loaded by pulling. This enables simple knotting at the end disks and prevents undesired induced bending moments when pushing the tendons.

We consider braided threads made of Nanofil® (ø 0.1 mm), Fireline® (ø 0.2 mm), Dyneema® (ø 0.14 mm), or Spectra Fiber® (ø 0.13 mm). To select a suitable thread

material, we experimentally determine the thread with the highest Young modulus in a tensile test, which is defined as the ratio of the stress (in gigapascals) to the strain (as a percentage). The stress is defined as the force divided by the individual cross-section of each thread. The tensile strain is defined as the elongation divided by the nominal thread length. We wind up each tendon on a motor shaft and stretch the thread at a constant rate until it breaks. The force is measured using a radial force sensor (RFS® 100, Checkline Europe, Bentheim, Germany). We pretension the tendons to 0.5 N before stretching them. The stretching rate is set at 0.014 mm/s. The thread length is fixed at 240 mm. Please note that this tensile test procedure to determine the Young modulus is not standard, but considered sufficient for comparison of the four braided threads.

The stress–stain diagrams shown in Figure 5 summarize our results. We can observe that Spectra Fiber® and Nanofil® has the highest Young modulus, about 285 GPa. This value deviates greatly from the manufacturer's information (about 120 GPa). We assume that this deviation is caused by the non-standard execution of the tensile test. The linear relationship between the strain and stress is positive for the modeling. In comparison, Spectra Fiber® shows constant linear behavior until it breaks. Nanofil®, however, experiences a partial failure at around 1% strain, indicating its lower resistance against sharp edges, which is also stated by the manufacturer. Based on this investigation, we use Spectra Fiber® to actuate the robot prototype in this paper.

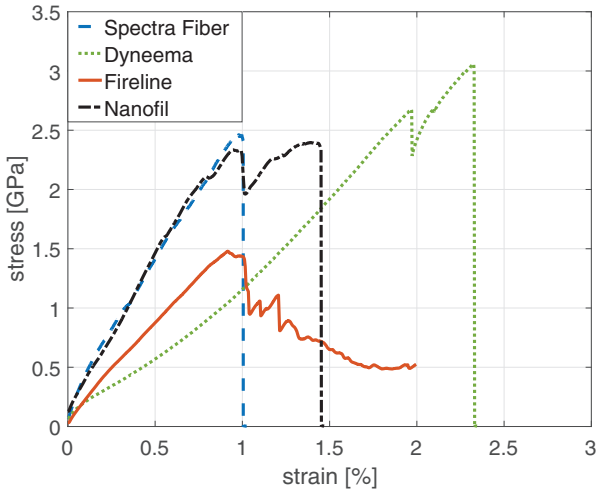


Fig. 5. Stress–strain curves of tensile tests for braided tendons made of Spectra Fiber®, Dyneema®, Fireline®, and Nanofil®.

2.4. Tendon elongation modeling

Although Spectra Fiber® shows low strain under load, the superposition of strains in a three-section robot can lead to a higher overall shape error. Therefore, a prediction of the tendon elongation behavior under applied loads is essential, in order to compensate for it.

Generally, there are three different elongation types in braided threads: elastic elongation, permanent elongation, and hysteresis. Elastic elongation can be immediately recovered after releasing the load, permanent elongation remains as permanent elongation after applying a load, and hysteresis recovers slowly over time. This behavior is based on the braided structure of the threads. Under load, the fibers rearrange until they reach an energy minimum.

To demonstrate its effects, we stretch a 240 mm long tendon for a defined duration (experimental setup similar to the tensile test described), and observe the force progression over 5 min in the stretched state. Beforehand, we pretension the tendon to 0.5 N with 0.014 mm/s. After 5 min, we return the tendon to its original position. We repeat this procedure five times. As the red dotted lines in Figure 6(top) show, we obtain a linear relationship during the constant-rate stretching, which is also observed in the Young modulus investigation (compare with Figure 5). However, after the stretching stops at around 100 s, the tendon force decreases over time and converges to a stable force value. After 5 min, the value changes become insignificant. The force change over time is the result of hysteresis, which describes the elongation of the tendon over time and, in the reverse case, the decrease in the tendon force.

The second effect we can observe in Figure 6(top), is a decreasing final force after repeated stretching. This behavior is the impact of permanent elongation, which occurs by loading the thread with similar loads or higher loads than the thread was exposed to before. Another evidence of permanent elongation is the decreasing pretension force at the beginning of the measurement (at time 0 s). It is reported that permanent elongation did not occur when the thread was prestretched to higher loads than expected in a desired application. Thus, we prestretch the thread with 30 N several times. After ten times, we could achieve repeatable results. The thread is tensioned to 0.5 N at the beginning of each prestretching repetition, at a rate of 0.014 mm/s. As the blue dotted lines in Figure 6(top) show, the force progression can be repeated with high accuracy.

To establish a model describing the relationship between the tendon force and the resulting elongation, we performed the procedure for obtaining the blue force progression from Figure 6(top) for several stretching distances. We chose four distances, resulting in the red lines in Figure 6(middle). We captured force progressions of a further four stretching distances, shown in blue lines in Figure 6(middle) for validating the tendon force model. The repeatability of the captured measurements is 0.1 N in the worst case and is defined as the difference between the maximum and minimum force value at 400 s over five repetitions. We interpolated a stress–strain curve based on the measured points, to be used for the modeling as a linear function.

The model shows almost linear behavior (see Figure 6(bottom), blue dashed line). According to the model, the diamonds mark the force value of the validation points. The red circles represent the measured validation points. The model and the validation match with a maximum deviation

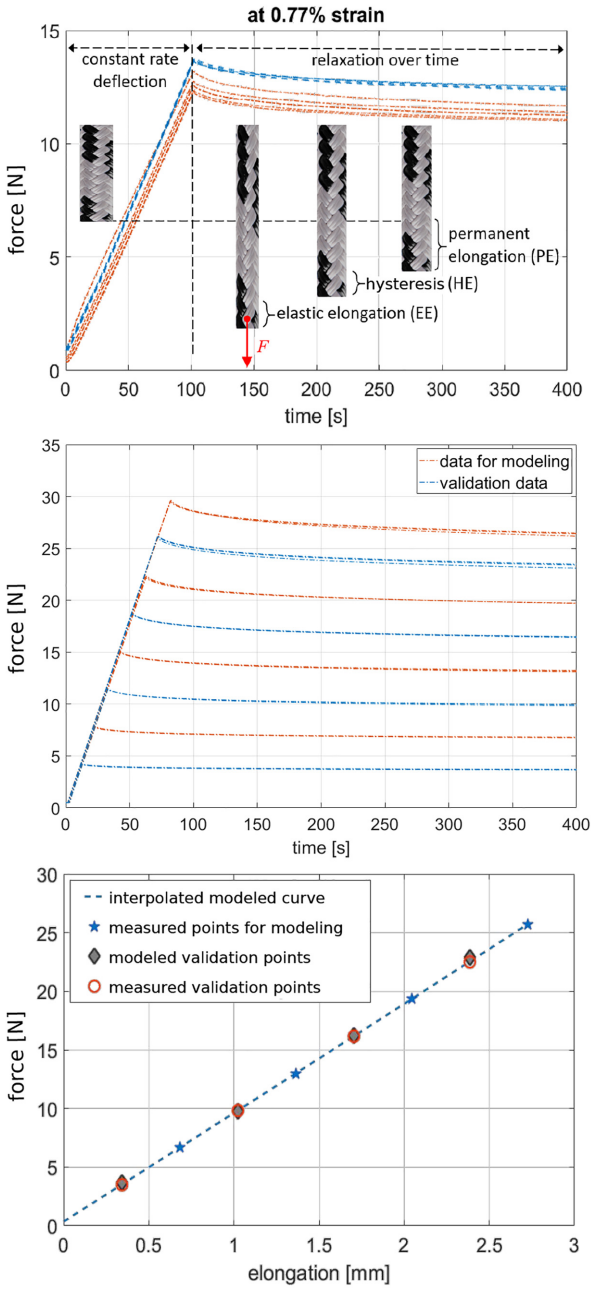


Fig. 6. Top: General behavior of tendons with permanent and elastic elongation, and hysteresis explained through force measurement over time. Red dotted lines: non-prestretched tendon measurements; blue dotted lines: prestretched tendon measurements. The relative elongation 0.77 % is in respect to 240 mm overall tendon probe length. Middle: Force over time for different elongations. Red: data used to derive the model. Blue: data used to validate the model. Bottom: Tendon model accuracy based on model and validation data.

of 3% of the respective force value. The function describing the model is used in the experimental evaluation section to obtain the tendon elongation dependent on the force

$$\Delta t(F) = 0.11F - 0.0383. \tag{3}$$

3. Robot kinematics

The forward kinematic model is based on Kirchhoff's rod theory, being a special case of Cosserat rod theory with the assumptions of (1) inextensibility and no transverse shear and (2) linear constitutive equations for bending and torsion. These assumptions have been proven efficient in Rucker et al. (2010) and Rucker and Webster, III (2011). Furthermore, we neglect any effects from friction. The model was previously presented in Neumann and Burgner-Kahrs (2016). Here, we state the changes and definitions required to follow this paper. We also refer the interested reader to a comparison of modeling approaches for extensible-section tendon-actuated continuum robots in Chikhaoui et al. (2019). In extension of Neumann and Burgner-Kahrs (2016), we show how to derive the tendon length displacements from the kinematics and apply the tendon force elongation model.

3.1. Forward kinematics

A robot is composed of n sections, where $i = 1$ denotes the proximal section and $i = n$ denotes the most distal section. The curve $\boldsymbol{g}(s) \in SE(3)$, describing the centerline position $\boldsymbol{p}(s) \in \boldsymbol{R}^3$ and material orientation $\boldsymbol{R}(s) \in SO(3)$, of the backbone in respect to arc length s is given as

$$\boldsymbol{g}(s) = \begin{bmatrix} \boldsymbol{R}(s) & \boldsymbol{p}(s) \\ \boldsymbol{0}^{\mathrm{T}} & 1 \end{bmatrix}, \tag{4}$$

with $s \in [0, l]$ and $l = \sum_1^n l_i$ the overall length of the robot. The linear and angular rates of change of $\boldsymbol{g}(s)$ with respect to arc length are represented by the kinematic variables $\boldsymbol{v}(s)$ and $\boldsymbol{u}(s)$. The evolution of $\boldsymbol{g}(s)$ is then defined as

$$\begin{aligned} \dot{\boldsymbol{R}}(s) &= \boldsymbol{R}(s)\hat{\boldsymbol{u}}(\boldsymbol{s}) \\ \dot{\boldsymbol{p}}(s) &= \boldsymbol{R}(s)\boldsymbol{v}(\boldsymbol{s}). \end{aligned} \tag{5}$$

The $\hat{}$ operator indicates transformation of a vector in $\mathbb{R}^3$ into a skew symmetric matrix in $\mathfrak{so}(3)$, and $\cdot$ denotes the derivative with respect to s. A robot configuration is defined as $\boldsymbol{q} = [l_1, l_2, l_3, F_{11}, F_{12}, F_{13}, F_{21}, F_{22}, F_{23}, F_{31}, F_{32}, F_{33}]$, with l_i the segment lengths and F_{it} the tendon forces. Figure 7 illustrates the kinematic variables.

The moment and force balance is defined by constitutive laws as stated in Neumann and Burgner-Kahrs (2016) and leads to the explicit forward kinematic model equations

$$\dot{\boldsymbol{v}} = -\boldsymbol{K}_{\mathrm{se}}^{-1}\left(\hat{\boldsymbol{u}}\boldsymbol{K}_{\mathrm{se}}(\boldsymbol{v} - \boldsymbol{e}_3) + \boldsymbol{R}^{\mathrm{T}}\boldsymbol{f}\right), \tag{6}$$

$$\dot{\boldsymbol{u}} = -\boldsymbol{K}_{\mathrm{bt}}^{-1}\left(\hat{\boldsymbol{u}}\boldsymbol{K}_{\mathrm{bt}}\boldsymbol{u} + \hat{\boldsymbol{v}}\boldsymbol{K}_{\mathrm{se}}(\boldsymbol{v} - \boldsymbol{e}_3) + \boldsymbol{R}^{\mathrm{T}}\boldsymbol{l}\right), \tag{7}$$

with the stiffness matrix of shear and extension $\boldsymbol{K}_{\mathrm{se}}$, the stiffness matrix for bending and torsion $\boldsymbol{K}_{\mathrm{bt}}$, distributed forces $\boldsymbol{f}$ acting on the backbone, and applied moment distribution $\boldsymbol{l}$. Initial conditions are adopted from Neumann and Burgner-Kahrs (2016). In this paper, the boundary

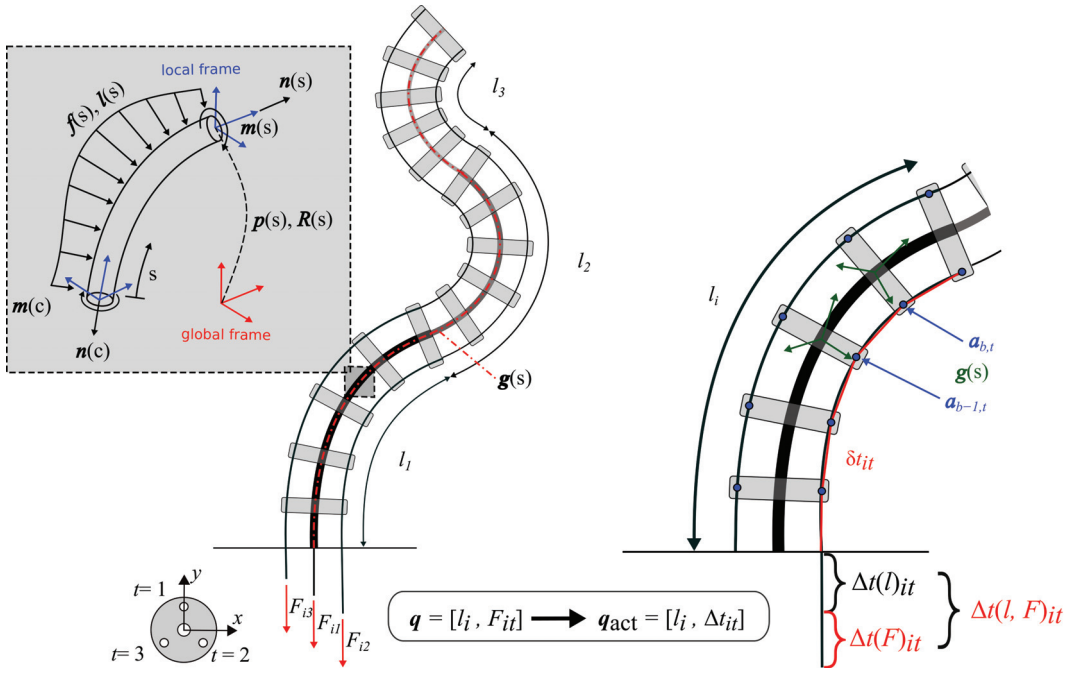


Fig. 7. Left: Kinematic input values with section lengths l_i and tendon forces F_{it}, illustrated for a schematic robot structure. Indication of kinematic approach with local frames along the robot structure ($\boldsymbol{p}(s)$, $\boldsymbol{R}(s)$), forces ($\boldsymbol{f}(s), \boldsymbol{n}(s)$) and moments ($\boldsymbol{l}(s)$, $\boldsymbol{m}(s)$) equilibrium. Right: Actuation input values $\Delta t(l, F)_{it}$ calculation on a schematic bending through a single tendon, considering tendon elongation due to load.

conditions are defined as moments at respective end disk locations induced through the tendon forces F_{it} with a lever arm r_t, such that for the first section, forces F_{1t} are considered. An ordinary differential equation solver is used to determine the solution to the equations for each section of the robot, starting with the proximal section. A root-finding solver is applied at intermediate steps to guess new initial values for $\boldsymbol{u}(s)$.

3.2. Tendon displacement and elongation calculation

To actuate the prototype, tendon length displacements are used, such that a conversion from $\boldsymbol{q}$ to $\boldsymbol{q}_{\text{act}} = [l_1, l_2, l_3, \Delta t_{11}, \Delta t_{12}, \Delta t_{13}, \Delta t_{21}, \Delta t_{22}, \Delta t_{23}, \Delta t_{31}, \Delta t_{32}, \Delta t_{33}]$ is necessary (see Figure 7). To determine the tendon displacements Δt_{it}, we accumulate the straight shortest distances δt_{it} between two routing channels of neighboring spacer disks, as shown in Figure 7(right) and subtract them from the corresponding section lengths l_i, as described in equation (8). The location of the routing channel $\boldsymbol{a}_{b,t}$ is determined by the center of the backbone $\boldsymbol{g}(s)$ and the respective tendon-routing position $\boldsymbol{c}_t$ on the disk

$$\boldsymbol{a}_{b,t} = \boldsymbol{c}_t \boldsymbol{g}(s)$$

where

$$\boldsymbol{c}_1 = [0 \;\; \mathrm{r_t} \;\; 0 \;\; 1]^{\mathrm{T}},$$

$$\boldsymbol{c}_2 = [r_t \cos(30°) \;\; -r_t \sin(30°) \; 0 \;\; 1]^{\mathrm{T}},$$

$$\boldsymbol{c}_3 = [-r_t \cos(30°) \;\; -r_t \sin(30°) \; 0 \;\; 1]^{\mathrm{T}}.$$

To account for the tendon elongation due to stress, we integrate the elongation part to the loaded tendons derived from the tendon elongation model (equation (3)), resulting in the overall equations

$$\Delta t(l)_{it} = -\left(l_i - \sum_{1}^{p \cdot i} \delta t_{it}\right) \tag{8}$$

$$\Delta t(l, F)_{it} = -\left(l_i - \sum_{1}^{p \cdot i} \delta t_{it}\right) + \Delta t(F)_{it}. \tag{9}$$

4. Theoretical evaluation

In this section, we investigate the workspace and the path-following motion in simulation of the proposed robot design. The following robot parameters and intervals for configurations are used: $l_i \in [25 \ldots 55\ \text{mm}]$, $F_{1t} \in [0 \ldots 30\ \text{N}]$, $F_{2t} \in [0 \ldots 20\ \text{N}]$, and $F_{3t} \in [0 \ldots 5\ \text{N}]$ with $i/t \in [1 \ldots 3]$. The minimum section length is chosen such that the section can still bend. The tendon force F_{it} ranges are chosen based on geometrical and material properties of the respective backbones, such that self-collision of the sections is avoided.

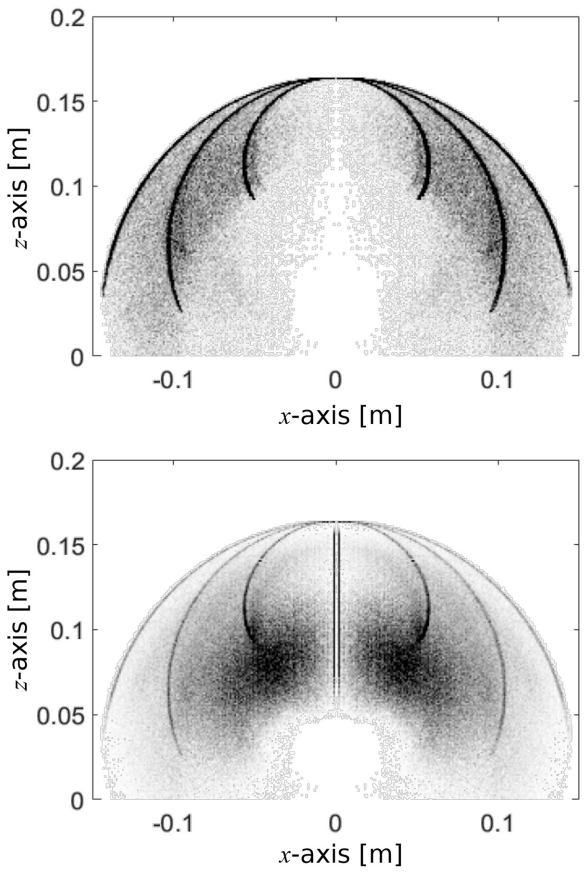


Fig. 8. Workspace projection on x–z-plane. Top: Fixed-sections tendon-driven robot. Bottom: Extensible-sections tendon-driven robot. End-effector positions are illustrated in a plane. The gray scales indicate the position redundancy, darker shades correspond to higher redundancies.

4.1. Workspace

Here, we seek to determine the workspace increase for a tendon-driven continuum robot with extensible sections compared with a tendon-driven continuum robot with fixed section lengths. We sample 5 million random configuration values within the intervals stated. For the fixed-section-length robot, we only vary the tendon forces and use a segment length of 55 mm. Tube parameters are chosen as in Table 1. The workspace is described by all reachable end-effector positions for the sampled configurations. All end-effector positions are projected into the x–z-plane and mirrored about the z-axis to account for the rotational symmetry of the workspace (see Figure 8).

To determine the workspace volume, we use the voxelization method proposed by Burgner-Kahrs et al. (2014) with isotropic voxels of 1 mm^3. Within the x–z-plane, all occupied voxels correspond to a volume of 35.3 cm^3 for the extensible-section robot in comparison with 28.8 cm^3 for the fixed-section-length robot. This corresponds to an enlargement of 22.5 %, which indicates the same relative increase for the 3D workspace volume. The workspace enlargement can particularly be observed in the center of the workspace. The shade of gray of each voxel in Figure 8 corresponds to the number of end-effector positions in this unit volume. The darker the shade, the more configurations lead to the same end-effector position, which is an indicator of the position redundancy of the robot. As the configuration space of the fixed-section-length robot is reduced, 5 million samples lead to a higher position redundancy (Figure 8(top)) in direct comparison with the extensible-section robot (Figure 8(bottom)). In fact, the position redundancy in the overlapping workspace areas of both robots is similar in reality. However, we want to point out that the variable-section-length robot also shows increased position redundancy in the center of the workspace, whereas a fixed-section-length robot undergoes a natural loss in redundancy toward the boundary of its workspace—typically in the center of the workspace.

4.2. Path-following motion

In this investigation, we regard the problem of path-following motion of an extensible continuum robot along a desired path. The desired path is represented as a general 3D curve $\boldsymbol{h}(s) \in SE(3)$, describing the position of the path and its orientation (Bishop frame) as a function of arc length s in task space coordinates. The origin $\boldsymbol{h}(0)$ coincides with the origin of the robot coordinate system. We assume that the length of $\boldsymbol{h}(s)$ is less than or equal to the maximum length of the extensible continuum robot. We refer to $\boldsymbol{h}(s)$ as the reference path in the following, as introduced in Neumann and Burgner-Kahrs (2016). Without loss of generality, we describe the path-following motion strategy for extensible continuum robots with $n = 3$ sections and note that the method is generalizable to general n. As the extensible continuum robot exhibits approximately constant curvature per section, we consider paths composed of three concatenated constant curvature segments seg_j (see Figure 9a). Each segment is defined by its length l_{seg_j}, segment curvature κ_{seg_j}, and segment angle ϕ_{seg_j}.

4.2.1. Strategy. We define the start configuration of the robot such that all sections are at their minimum length while conforming to the initial curvature of the reference path $\boldsymbol{h}(s)$. The path-following motion strategy extends each section sequentially, starting from section $i = 1$ in equidistant steps δ. The length of section 1 is extended step-wise by δ until the length of seg_1 is reached. Then, section 2 is extended until it reaches the length of seg_2, and so on. During extension of a section, we allow the succeeding sections to adapt their curvatures such that they conform to the curvature of the path. A section i adopts to a segment j that contains more than 50% of the section length.

Since the section lengths l_i are set by the deployment strategy, only the tendon forces need to be determined as inputs to the kinematics. Therefore, we perform numerical

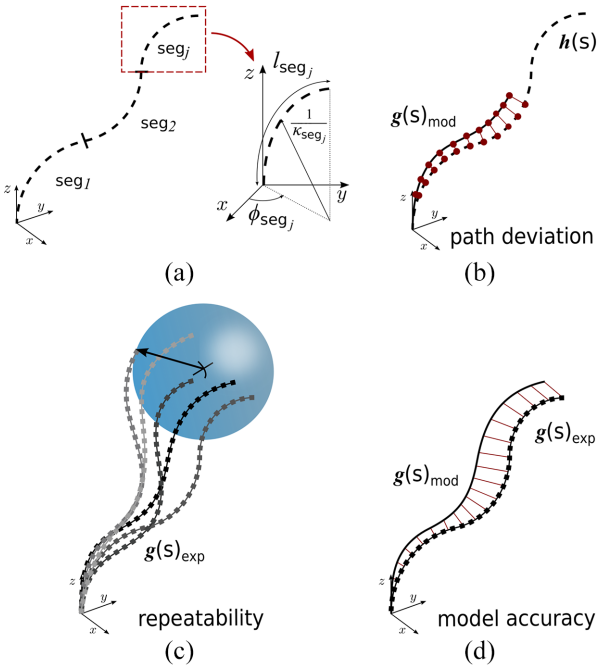


Fig. 9. (a) Desired path definition with the parameters of an arc. Error definitions: (b) path deviation, (c) repeatability, and (d) model accuracy.

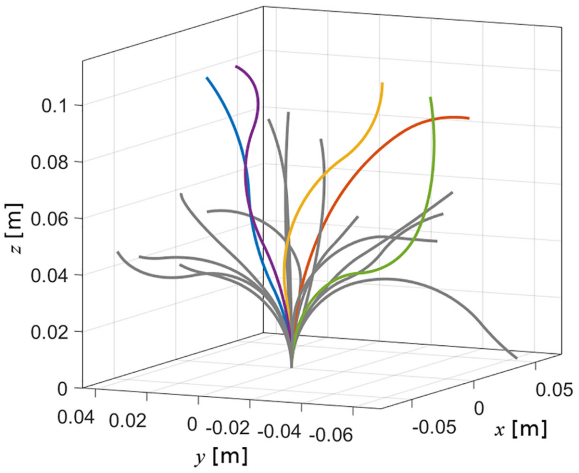


Fig. 10. Evaluated paths for path-following deployment. All paths used for the theoretical investigation; colored paths also used for the experimental evaluation.

determination of roots, applying sequential quadratic programming for each deployment step, and use the forward kinematics with inputs F_{it} and l_i for $i/t \in [1 \dots 3]$, while only tendon forces are varied. The objective function to minimize is the path deviation error (see next section) for the current deployment step. We allow a maximum of two tendons per section to be tensioned at the same time, to prevent backbone buckling. For the numerical search, thus results in six values that are optimized.

4.2.2. Error metric. The path deviation error is defined as the Euclidean distance between corresponding points of the simulated backbone $\boldsymbol{g}(s)_{\text{mod}}$ and reference path $\boldsymbol{h}(s)$, as shown in Figure 9(b), with $s \in [0 \dots \sum_{i=1}^{3} l_i]$. Both curves are discretized in 100 equidistant points. We do not distinguish between tip error and path deviation error, since it has been shown that there is no positive effect in considering several error types in the objective function (Neumann and Burgner-Kahrs, 2016). Note that the maximum path deviation is not always at the tip of the backbone.

4.2.3. Initialization. Depending on the angle ϕ_{seg_j} of the reference path, we can predict which two of three tendons in the sections relevant for path segment j will be actuated. For each deployment step, we define beforehand which tendons must be optimized. The starting force values for these tendons are generated randomly as follows: $F_{1t} \in [0 \dots 30\text{N}]$, $F_{2t} \in [0 \dots 20\text{N}]$, and $F_{3t} \in [0 \dots 5\text{N}]$. Those values are chosen such that the curvatures do not exceed the material strain and the robot does not self-collide. The minimum section lengths are predefined with respect to the maximum curvature $\max(\kappa_{\text{seg}_j})$ of the reference path and are determined by

$$l_{i,\,\min} = \frac{l_{\min}}{1 - \mathrm{r_d} \cdot \max(\kappa_{\text{seg}_j})}, \tag{10}$$

4.2.4. Execution and results. We evaluate the path-following deployment for an extensible continuum robot with the parameters stated in Table 1. We generate 25 random paths with parameters $\kappa_{\text{seg}_j} \in [0 \dots 35^{-m}]$, $\phi_{\text{seg}_j} \in [0 \dots 2\pi]$, and $l_i \in [l_{\min}, l_{\max}]$, such that kinematic constraints are fulfilled. The paths are depicted in Figure 10.

The step size is set to $\delta = 0.5$mm. Figure 11 shows selected deployment steps for an example path. The reference path is depicted in red and the path deviations between the robot's backbone and the reference path are illustrated as projections on each coordinate plane. It can be observed that the path deviation error is larger at the transitions of the path curvature. For all 25 paths, the maximum path deviation errors are less than 5 mm and the median path deviation error is less than 1.5 mm, as depicted in Figure 12. Each box plot includes the path deviation errors of all deployment steps for a single path. The boxes define the 25th and 75th percentiles. Outliers are values higher than 1.5 times the interquartile range. The errors are calculated relative to the corresponding robot length in each current configuration. If the numerical search does not result in smaller path deviation error than $e_{\max} = 5$mm after 1000 iterations (owing to converging to a local minimum), we repeat the search with a new force values guess, as mentioned in the previous section. The threshold $e_{\max}$ is chosen based on Neumann and Burgner-Kahrs (2016). Higher accuracy is achievable by setting the maximum error

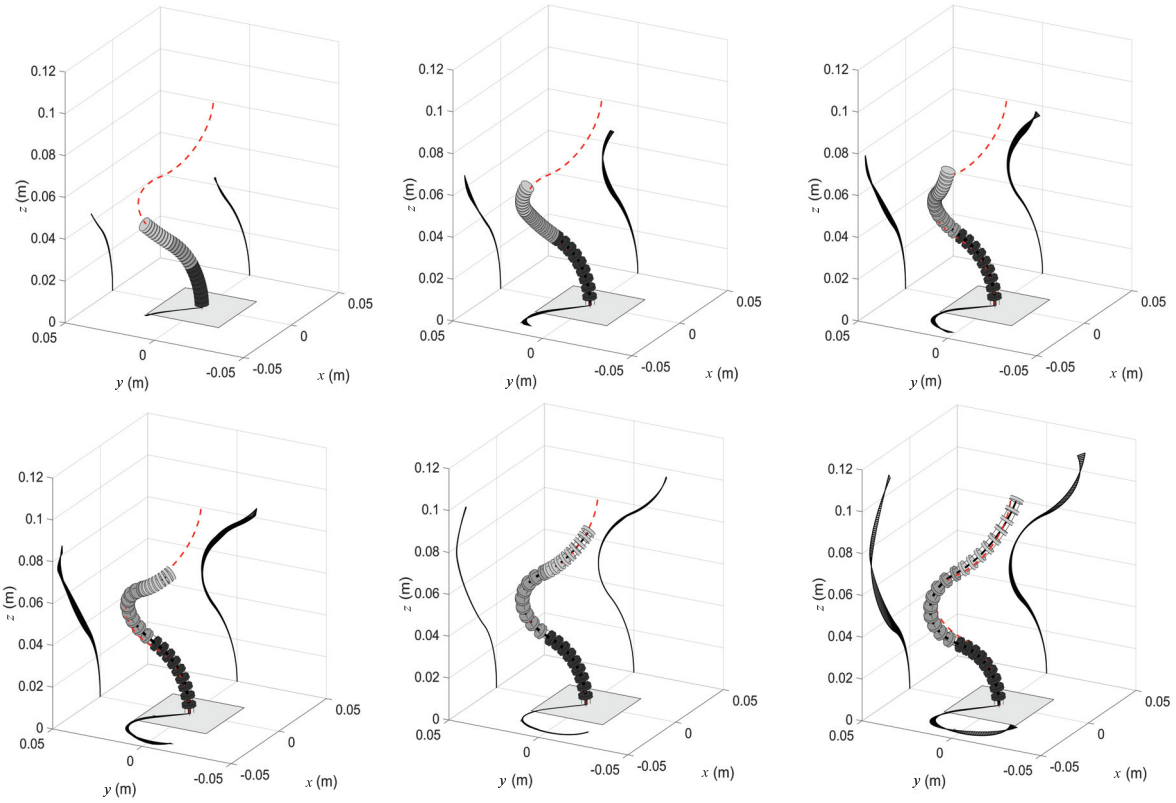


Fig. 11. Simulated deployment along a reference path with three constant curvature arcs (red dashed line) and path deviation errors as shadows on each plane. Six example deployment steps.

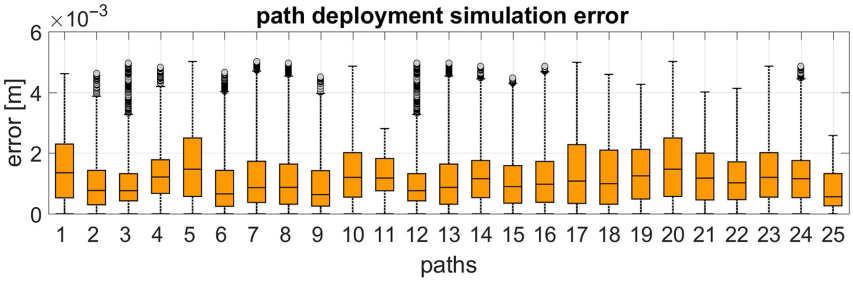


Fig. 12. Path deviation errors of 25 paths. Each box plot contains all intermediate path deviations for one path.

boundary lower, as we will show in the experimental path-following motion evaluation. On an Intel(R)Core(TM) i7-3770 CPU (3.4 GHz, four cores), a shape optimization run for a single configuration is performed in 793 s (349–1068 s). The computation efficiency is strongly related to the initial force guess values. The overall time for a deployment sequence is governed by the maximum acceptable error allowed.

5. Experimental evaluation

For the experimental investigations, we use the robot and actuation unit prototype previously described. The measurements are performed using a scan arm (Faro Edge Scan Arm, Faro Europe, Korntal-Münchingen, Germany), which allows tactile coordinate measurements and non-contact spatial laser scans (precision with maximum error of 0.041 mm). The experimental setup is depicted in Figure 13. The configuration parameter intervals are $l_i \in [25 \dots 55]$ mm and $F_{1t} \in [0 \dots 30\text{N}]$, $F_{2t} \in [0 \dots 20\text{N}]$, and $F_{3t} \in [0 \dots 5\text{N}]$, with $i/t \in [1 \dots 3]$ (similar to the previous section).

5.1. Repeatability

To determine the repeatability of our robot prototype, we consider 40 randomly sampled configurations (Figure 14(right)). The robot is driven into each of these

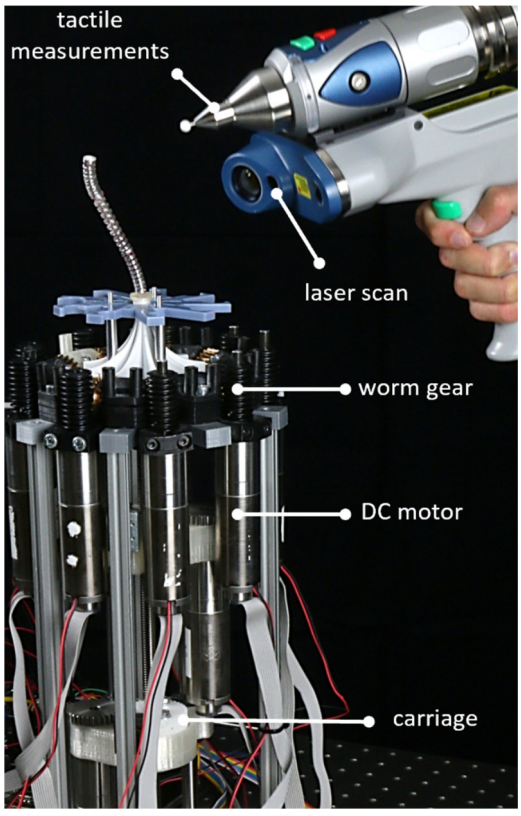


Fig. 13. Experimental setup, showing the actuation unit prototype and the measuring arm with a tactile probe and the laser scanner.

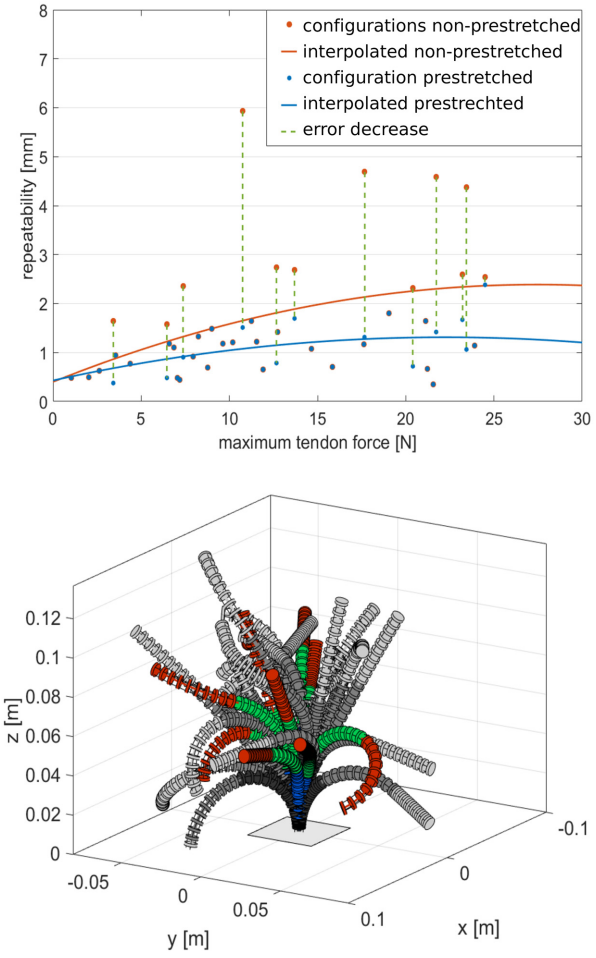


Fig. 14. Top: Repeatability for 40 random robot configurations. Horizontal axis indicates the maximum tendon force of the overall configuration. Orange dots for non-prestretched tendon measurements, blue dots for prestretched tendon measurements. Solid lines illustrate interpolated repeatability. Green dashed lines show the repeatability improvement. Bottom: Configurations used for repeatability investigations (all colors), colored configurations are used for model tuning; 25 of the gray configurations are used for model accuracy investigations.

configurations from its initial straight configuration and the position of the tip is measured using the coordinate measurement system. Each configuration is approached five times. Before driving the robot into a configuration, the pretension of the tendons is manually set to 0.5 N, using a spring scale. The repeatability is defined by the radius of a sphere containing all five measured tip positions (see Figure 9(c)). The center of the sphere is the mean position of five measurements for the same configuration. The largest Euclidean distance of the tip positions to the sphere center defines the radius.

To compare the effect of tendon prestretching, the repeatability is determined for non-prestretched and prestretched tendons. To prestretch the tendons in the actual prototype, we drive the robot by a single tendon 10 times with maximum tendon displacements. This procedure is performed for all nine tendons. The results of the non-prestretched tendons are shown in orange in Figure 14(top).

We sort the repeatability values over tendon forces to show the correlation. Additionally, the orange solid curve shows the fitted quadratic function. It can be observed that the repeatability increases with higher tendon forces. The repeatability in the non-prestretched experiments ranges between 0.35 and 6 mm, with a median of 1.22 mm. We repeat the measurements with the prestretched tendons for 11 configurations with repeatabilities exceeding the standard deviation in order to show the effect of prestretching on some examples. The results are depicted in blue dots and show repeatability between 0.35 and 2.38 mm, with a median of 1.08 mm. The corresponding repeatability values for non-prestretched and prestretched tendons are connected by green dashed lines to show the improvement. The interpolation curve for prestretched experiments shows lower magnitudes, but a correlation between the tendon force and the repeatability can still be observed. The repeatability may be further influenced by mechanical play, manufacturing tolerances, and uncompensated tendon elongation.

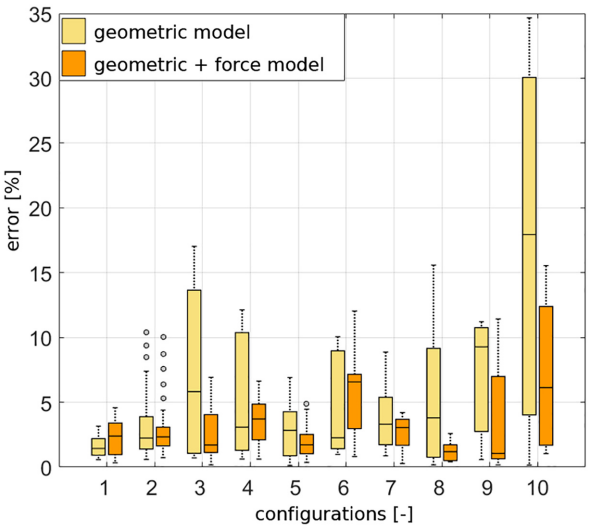


Fig. 15. Model accuracy error for 10 configurations with geometric (yellow) and combined (orange) tendon displacement calculation approach. The errors are calculated relative to the corresponding configuration robot lengths.

5.2. Model accuracy

To evaluate the model accuracy, we compare the calculated robot shape using the kinematic model with the prototype shape established by the tendon displacement actuation. The calculated shapes are selected from the configuration pool (tendon forces F_{it} and segment lengths l_i) (Figure 14 (bottom), colored configurations). To actuate the robot prototype, we use both the tendon displacement calculation approach based solely on geometry and the approach based on geometry and tendon forces.

The prestretching and the pretension were performed in the same manner as described for the repeatability experiments. Each configuration shape is measured using the laser scan arm, which results in a 3D point cloud defining positions on the surface of the robot. The scan arm was manually operated. The centerline is extracted from the measured point cloud using the L1 algorithm introduced by Huang et al. (2013). The algorithm requires an equal distribution of cloud points to determine the centerline reliably. Therefore, the point cloud should include all the surfaces of the robot. For equal distribution, the point cloud is reorganized and filtered to a grid using the manufacturer's software. To compare the modeled and measured robot shapes, we discretize the modeled and measured centerlines into 100 equidistant points. The model error is then defined as the mean over all Euclidean distances between corresponding points (see Figure 9(d)).

5.2.1. Model tuning. To tune the model parameters, we use 10 randomly chosen configurations (Figure 14(bottom)). We calculate the robot shape using the kinematic model for each configuration, with the parameters stated in Table 1. The robot then moves into each configuration. The measurement and the centerline extraction is performed as described for the repeatability experiments. Having the corresponding calculated and measured robot shapes, we minimize the shape deviation by optimizing the outer diameters of the three tubes. We chose these parameters, since the manufacturing process of the tubes does indeed come with some tolerances. Furthermore, the diameters of the tubes can compensate for unknown material properties, such as the Young modulus. The exact value of the Young modulus is not provided by the manufacturer and can vary from tube to tube. We define the optimization problem as a constraint minimization of the shape deviation, as described in the path-following motion strategy previously outlined. The diameters are constrained to 1 ± 0.04 mm, 0.8 ± 0.04 mm, and 0.5 ± 0.04 mm, owing to the production tolerances of

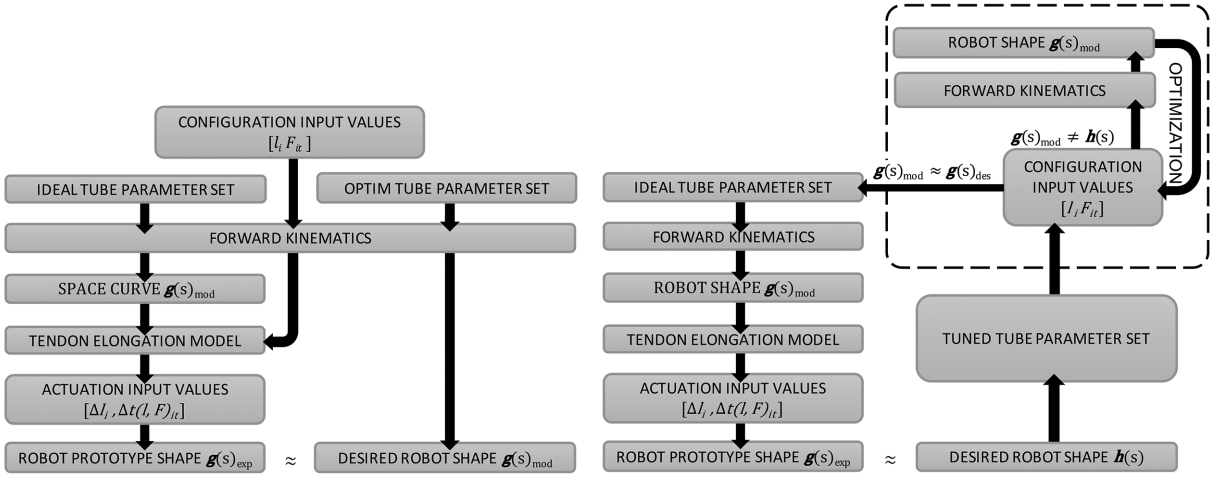


Fig. 16. Left: Workflow for experimental evaluation of the model accuracy. Right: Workflow for experimental evaluation of the path-following evaluation.

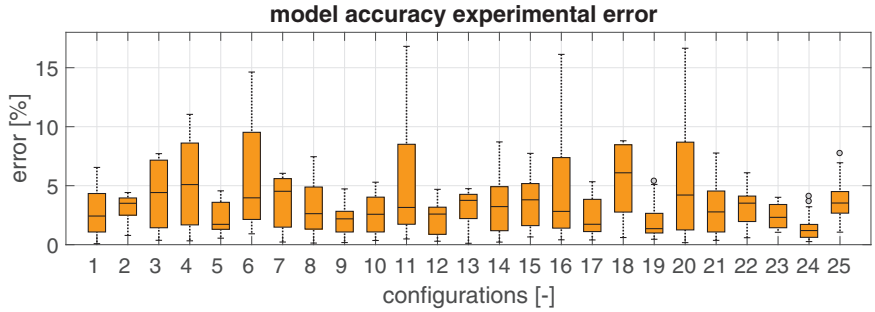


Fig. 17. Model accuracy errors for 25 configurations. Errors are calculated relative to the corresponding robot lengths.

the extrusion processes and the inner and outer diameters of the next outer tube. Sequential quadratic programming is used for optimization. The optimization converged to 1.037 mm for the outer tube, 0.797 mm for the middle tube, and 0.54 mm for the inner tube at a median modeling error of 3.21% (3.81% without the tendon elongation model). The model accuracy errors over 10 configurations are depicted in Figure 15. Each box plot pair (orange and yellow boxes) includes the errors of the same configuration. The boxes include the 75th and 25th percentiles with outlier values exceeding 1.5 times the interquartile range. The errors are calculated relative to the corresponding configuration robot length.

5.2.2. Accuracy investigations. For model accuracy investigations, we extend the results from the model tuning by evaluating an additional 25 configurations. The configurations are chosen randomly from the configuration pool in Figure 14(bottom) (colored gray). The workflow is shown in Figure 16(left). Using the tuned model parameters, the expected robot shape is determined as $\boldsymbol{g}(s)_{\text{mod}}$. The corresponding actuator inputs for the robot prototype are calculated using the kinematic model compensating for tendon elongation. The robot then moves into the configuration, the shape is measured using the laser scan arm, and the centerline is extracted ($\boldsymbol{g}(s)_{\text{exp}}$). The shape deviation between $\boldsymbol{g}(s)_{\text{mod}}$ and $\boldsymbol{g}(s)_{\text{exp}}$ is measured, as described previously. Figure 17 shows the model accuracy errors of 25 measured configurations. Each box plot represents the shape deviations observed for a single configuration. The relative deviation is reported, in terms of the robot length of the respective configuration. The boxes include the 75th and 25th percentiles, with outlier values exceeding 1.5 times the interquartile range.

The majority of the errors over all configurations (75%) lies below 10%. The overall median error of 3.16% is similar to the tuning results. Some configurations show larger maximum deviations, up to 17% of the robot length. It can be observed that these configurations are characterized by two actuated tendons in the same section, with one relatively low force (less than 2 N). We assume that this behavior is caused by tendon friction, erasing lower tendon forces and therefore the bending associated with this forces. The impact of friction on the model accuracy requires further investigations, which are the subject of future work. In addition to friction, mechanical play and manufacturing tolerances contribute to the repeatability at the end-effector up to about 2.38 mm and impacts the model accuracy.

5.3. Follow-the-leader

In this set of experiments, we seek to evaluate the accuracy of the extensible continuum robot prototype in path-following motion. Figure 16(right) shows the workflow of the evaluation. Five paths ($\boldsymbol{h}(s)$) are picked from the simulation described previously in terms of different shapes of small and large curvatures (depicted in color in Figure 10). Those paths correspond to path numbers 3, 4, 9, 11, and 17 from the simulation investigations (Figure 12). We follow the path-following motion strategy described previously to optimize for tendon forces leading to a maximum shape deviation error below 2 mm for each deployment step. The deployment step size is set to $\delta = 5\text{mm}$. In each deployment step, the expected robot shape $\boldsymbol{g}(s)_{\text{mod}}$ is calculated using the forward kinematics with tuned tube parameters. Once the path-following motion strategy is established, the tendon forces are translated into respective tendon displacements using the tendon elongation model. The robot prototype is then deployed along each of the five paths. The shape measurements, centerline extraction, and calculation of the shape deviation error are performed for each deployment step, as described previously.

An example deployment for path 5 is shown in Figure 18 and the results for all five paths are summarized in Figure 19. Each box plot represents the errors over all deployment steps for the path. The shape deviation error stays below 6% (relative to the robot length) in 75% of the cases. The median error over all paths is 2.59%. The maximum error is 14%. The path deployment results are slightly better but comparable to the model investigation results, which is probably caused by the smaller number of chosen paths.

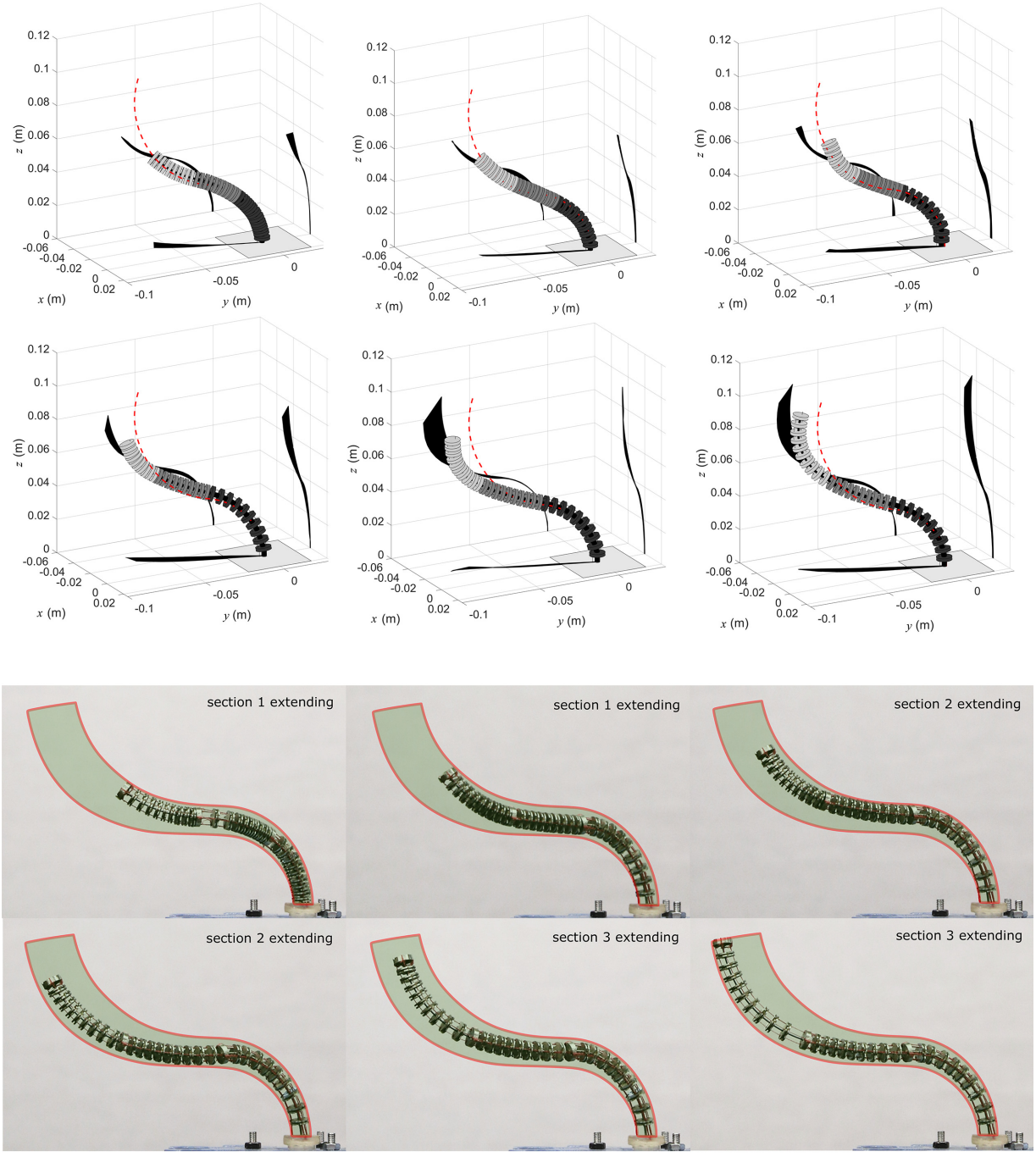


Fig. 18. Top: Experimental deployment sequence along a desired path with three constant curvature arcs (red dashed line) and path deviation errors as shadows on each plane. Bottom: Photographs of the deployment sequence ordered by the extending section. The overall occupied space over the course of deployment is marked in green.

6. Conclusion

We propose a tendon-driven continuum robot with extensible sections. Extensibility is achieved by a telescoping backbone structure and repelling magnetic tendon-routing disks in each section. A workspace volume increase of 22.5 % compared with tendon-driven continuum robots with fixed section lengths is achieved. Additionally, an increased position redundancy in the center of the workspace is observed.

We achieved a repeatability over 40 random configurations of maximum 2.38 mm for our robot prototype. The median model accuracy is 3.16 % in respect to robot length

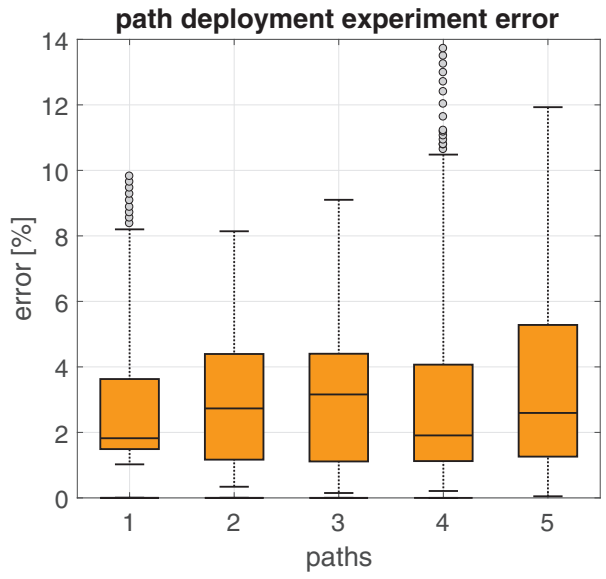


Fig. 19. Box plots of path deviation errors over all deployment steps of five paths. Errors (expressed as percentages) are relative to the corresponding configuration robot lengths.

over 25 measurements. For the first time, we assessed the path-following deviation of a tendon-driven continuum robot prototype and compared the results with simulations. The median deviation over five paths is 2.59% and the maximum path deviation is 14%, both in respect to the robot length. Both values show better results compared with the state of the art reported by Palmer et al. (2014).

As our robot design features segments bending with constant curvature, following general paths is not inherently feasible and may lead to larger path deviations (Neumann and Burgner-Kahrs, 2016). A continuum robot design to facilitate general path-following would require more degrees of freedom in the task space. As our results are obtained by model-based control, they may be significantly improved by implementing feedback control on the robot prototype (Chikhaoui and Burgner-Kahrs, 2018). This would involve the integration of sensing means for the tip pose, the section position, or, ultimately, the shape.

In this paper, the tendons of all sections are routed through the same routing channels. Friction between those tendons possibly negatively affects the accuracy. For future work, handling friction between the tendons and spacer disks is an important issue to address, either through modeling or mechanical adaptation. The friction between the tendons and the spacer disks is also the reason for slightly uneven spacer disk distributions for relatively high curvatures in the first and second segments (owing to the higher segment stiffness of the distal segment). Magnets with stronger magnetic fields may counter this behavior. This is particularly true for multiple-section continuum robots, as they involve simultaneous actuation of several tendons. Further, alternative deployment sequences strategies could be considered.

To conclude, the extensible tendon-driven continuum robot design presented in this paper shows good potential for applications in tortuous environments. The inherent deployment and path-following motion capabilities are beneficial for a variety of tasks.

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Note

1. Paper-related data can be shared with interested researchers on personal request.

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